Supplementary Methods and Computational Verification

Submission 6222. Revision extended 7 October 2026. This supplement records computed results and explicit limits. It contains no invented flow observation or field validation.

S1. Baseline definitions and source integrity

Main coverage baseline: geoBoundaries regional population mask; WorldPop or GHS-POP 2025; EPSG:6933 1-km cells; HOT road snapshot 5 October 2026; main classes; connectors up to 5 km; no deliberate deletion or hub displacement; 100-km coverage radius; fixed 26 existing candidate nodes. The original COD-AB WorldPop count is rechecked separately and does not replace the common-mask optimization results.

population_grid_metadata.json


```
{
  "status": "computed",
  "population_product": "WorldPop 2025 R2025A v1 constrained, alpha",
  "boundary_product": "geoBoundaries COD ADM1 gbOpen boundaryID COD-ADM1-28894549",
  "boundary_note": "alternative to COD-AB, do not interpret as identical administrative mask",
  "provinces": [
    "Kinshasa",
    "Kongo-Central",
    "Kwango",
    "Kwilu",
    "Mai-Ndombe",
    "Mongala",
    "Nord-Ubangi",
    "Sud-Ubangi",
    "Tshuapa",
    "Équateur"
  ],
  "crs": "EPSG:6933",
  "pixel_inclusion": "centre inside union of ten provinces; positive finite pixel counts only",
  "original_selected_pixel_sum": 43302131.740662836,
  "positive_source_pixels": 3651459,
  "positive_cells_1km": 219279,
  "positive_cells_5km": 24553,
  "mass_conservation_relative_error": 1.720603651003945e-16,
  "worldpop_sha256": "141ab3637e858a3597b96b7646dbf6b4c7d258a89a60a5ffcde07f051a109d5e",
  "boundary_sha256": "3deb46256765a4ce78a2e8dadaf6f93f43869877e55d242d1e68139b45a5395a"
}
```


population_product_comparison_metadata.json


```
{
  "worldpop_total": 43302131.74066284,
  "ghspop_total": 34033889.96602864,
  "ghspop_vs_worldpop_pct": -21.403661672228637,
  "positive_ghspop_cells_1km": 147956,
  "positive_ghspop_cells_5km": 21120,
  "cell_5km_pearson": 0.8713868250642843,
  "cell_5km_spearman": 0.9305105678697244,
  "sum_absolute_cell_difference": 23896670.506098725,
  "limits": [
    "both are modelled 2025 population products, not observations",
    "native GHS pixel centres allocated to EPSG6933 bins; grid alignment affects local comparisons",
    "same alternative regional boundary mask with pixel-centre inclusion",
    "GHS SMOD is modelled settlement classification; no functional autonomy inferred"
  ]
}
```

ROAD_ACCESSIBILITY_METADATA.JSON


```
{
  "snapshot": "2026-10-05",
  "source": "HOT/OSM HDX roads.gpkg",
  "population_mask": "geoBoundaries alternative",
  "lengths": "WGS84 ellipsoidal geodesic, km",
  "junctions": "exact shared source coordinates; interior layer key; endpoints level 0; no spatial snapping between roads",
  "direction": "oneway yes/true/1 forward, -1 reverse; all other values assumed bidirectional, including missing/reversible",
  "limits": [
    "no turn restrictions or access tags",
    "vertex attachment rather than nearest segment",
    "off-network straight connectors are hypothetical, not observed traversability",
    "missing layer/bridge tags may cause incorrect junctions",
    "roads selected by ten provincial pcodes; outside-province detours omitted",
    "no rail/fluvial routing or navigability validation",
    "26 hubs are existing urban nodes; this is not a new-hub optimization"
  ],
  "scenarios": [
    {
      "scenario": "road_classes",
      "classes": [
        "living_street",
        "primary",
        "primary_link",
        "residential",
        "road",
        "secondary",
        "secondary_link",
        "service",
        "tertiary",
        "tertiary_link",
        "trunk",
        "trunk_link",
        "unclassified"
      ],
      "oneway_nonempty_features": 1589,
      "oneway_yes_features": 1378,
      "layer_nonempty_features": 1353,
      "bridge_nonempty_features": 1332,
      "weak_component_sizes_top10": [
        1294595,
        187112,
        47697,
        37938,
        21865,
        18329,
        15393,
        12234,
        8893,
        4102
      ]
    },
    {
      "scenario": "including_tracks",
      "classes": [
        "living_street",
        "primary",
        "primary_link",
        "residential",
        "road",
        "secondary",
        "secondary_link",
        "service",
        "tertiary",
```

```
        "tertiary_link",
        "track",
        "trunk",
        "trunk_link",
        "unclassified"
      ],
      "oneway_nonempty_features": 1592,
      "oneway_yes_features": 1380,
      "layer_nonempty_features": 1420,
      "bridge_nonempty_features": 1398,
      "weak_component_sizes_top10": [
        1844769,
        300759,
        57381,
        45489,
        23308,
        20042,
        19192,
        16667,
        15659,
        5790
      ]
    }
  ]
}
```


JOINT_COVERAGE_METADATA.JSON


```
{
  "completed_scenarios_per_p": 96,
  "total_optimization_runs": 288,
  "objective": "maximize modelled population within 100 km road+assumed connector distance",
  "candidate_set": "26 existing GeoNames urban nodes, not proposed new cities",
  "factors": {
    "population": [
      "WorldPop2025",
      "GHSPOP2025"
    ],
    "grid_km": [
      1,
      5
    ],
    "tracks": [
      false,
      true
    ],
    "connector_km": [
      1,
      2.5,
      5
    ],
    "road_feature_removal_fraction": [
      0,
      0.01
    ],
    "hub_coordinate_displacement_km": [
      0,
      0.25
    ]
  },
  "perturbation_seed": 20261006,
  "removal_method": "one independent uniform draw per selected road feature; fixed sample for each graph scenario; not calibrated OSM incompleteness",
  "coordinate_method": "fixed 250 m displacement with independent random azimuth per hub; stress test not positional error
```

distribution",
  "limits": [
    "scenario ranges are not confidence intervals",
    "objective differs from legacy p-median",
    "alternative population masks and products are modelled",
    "equal-objective site sets may be nonunique; reported identities are one optimum",
    "not a Sobol decomposition",
    "no travel speeds because no travel-time claims",
    "no observed-flow validation"
  ]
}

EXACT_PMEDIAN_METADATA.JSON

{
  "demand_total": 7371525.750308432,
  "demand_definition": "sum of released 5-km node-buffer population estimates, not the regional population",
  "nodes": 26,
  "edges": 49,
  "distance": "released rounded edge distances in km",
  "solver": "scipy.optimize.milp / HiGHS"
}

CANDIDATE_WEIGHT_AUDIT_METADATA.JSON

{
  "candidate_count": 75,
  "models": [
    "reference seven criteria",
    "uniform seven criteria"
  ],
  "dirichlet_concentration": 60,
  "simulations_per_model": 5000,
  "seed": 20260902,
  "limits": [
    "only 75 previously sampled and filtered sites, not all eligible regional land",
    "one winner per province; no across-province separation constraint",
    "weights are assumed, not elicited expert judgments",
    "weight-only frequencies are conditional, not probabilities of feasibility",
    "protected areas, tenure and infrastructure capacity unverified"
  ]
}

S2. Boundary count comparison

COD-AB WorldPop selected count = 43,227,817.79641058. Common geoBoundaries count = 43,302,131.74066284. Difference = 74,313.94425226 (0.17191878% of COD-AB total). This comparison verifies the earlier total and is not a second regional optimization under COD-AB.

S3. All six road-reachability configurations

{"scenario": "road_classes", "max_connector_km": 1.0, "road_features": 106892, "graph_vertices": 1734033, "directed_arcs": 3594763, "weak_components": 1273, "eligible_hubs": 25, "population_total": 43302131.74066284, "population_attached_pct": 88.31512969232324, "population_reachable_pct": 78.4931613457577, "population_attached_no_hub_path_pct": 9.821968346565535, "population_unattached_pct": 11.68487030767675, "reachable_population_mean_distance_km": 76.58858441574219, "population_within100km_pct": 55.58750360632013}

{"scenario": "road_classes", "max_connector_km": 2.5, "road_features": 106892, "graph_vertices": 1734033, "directed_arcs": 3594763, "weak_components": 1273, "eligible_hubs": 26, "population_total": 43302131.74066284, "population_attached_pct": 92.51683193091556, "population_reachable_pct": 82.02619927898995, "population_attached_no_hub_path_pct": 10.4906326519256, "population_unattached_pct": 7.483168069084438, "reachable_population_mean_distance_km": 77.6537383250754, "population_within100km_pct": 58.10959650715915}

{"scenario": "road_classes", "max_connector_km": 5.0, "road_features": 106892, "graph_vertices": 1734033, "directed_arcs": 3594763, "weak_components": 1273, "eligible_hubs": 26, "population_total": 43302131.74066284, "population_attached_pct": 96.13577926081923,

"population_reachable_pct": 84.95129616127578, "population_attached_no_hub_path_pct": 11.184483099543453, "population_unattached_pct": 3.864220739180741, "reachable_population_mean_distance_km": 79.820525012691, "population_within100km_pct": 59.37470564073577}

{"scenario": "including_tracks", "max_connector_km": 1.0, "road_features": 147365, "graph_vertices": 2518218, "directed_arcs": 5187173, "weak_components": 2779, "eligible_hubs": 26, "population_total": 43302131.74066284, "population_attached_pct": 91.35801905655045, "population_reachable_pct": 81.43547394073144, "population_attached_no_hub_path_pct": 9.922545115819007, "population_unattached_pct": 8.641980943449537, "reachable_population_mean_distance_km": 76.02766316124873, "population_within100km_pct": 57.59358852077848}

{"scenario": "including_tracks", "max_connector_km": 2.5, "road_features": 147365, "graph_vertices": 2518218, "directed_arcs": 5187173, "weak_components": 2779, "eligible_hubs": 26, "population_total": 43302131.74066284, "population_attached_pct": 94.61709499687294, "population_reachable_pct": 84.16367984061533, "population_attached_no_hub_path_pct": 10.453415156257618, "population_unattached_pct": 5.382905003127044, "reachable_population_mean_distance_km": 77.43782279262729, "population_within100km_pct": 59.07054405524485}

{"scenario": "including_tracks", "max_connector_km": 5.0, "road_features": 147365, "graph_vertices": 2518218, "directed_arcs": 5187173, "weak_components": 2779, "eligible_hubs": 26, "population_total": 43302131.74066284, "population_attached_pct": 97.18304446656111, "population_reachable_pct": 86.1637486531681, "population_attached_no_hub_path_pct": 11.019295813393022, "population_unattached_pct": 2.816955533438859, "reachable_population_mean_distance_km": 79.12641692639068, "population_within100km_pct": 59.89509090659399}

S4. Complete audit tables and optimization runs

The replication package provides machine-readable topology metrics for all nodes and all graph variants; exact p-median outcomes for p = 2–20; 288 factorial and 108 additional repeated-stress maximal-covering result rows and selected-set rows; all candidate weight-centre comparisons; settlement classifications; population-product comparisons; and source metadata. These files are the numerical evidence underlying the manuscript tables. The plot code in build_final.py reads the same tables. A zero solver gap establishes an optimum for the encoded candidate set, not unique facilities or empirical correctness.

S5. Secondary observational source assessed, not used as validation

JICA, PDTK Final Report, April 2019, vol. 1, part 1: section 2.2.7, printed pp. 2-47–2-48, discusses GPS travel-speed surveys in 2017. Table 2.2.16 lists 13 Kinshasa urban routes. The source PDF SHA-256 is b0f5aa1a9b3f4845daa03afbbf8fe6347ade8ecf6e68902148bc38ea44108599. The retained route inventory is descriptive. No route matching, travel-speed digitization, or regional predictive validation has been performed. The report does not supply an OD target for the ten-province maximal-covering problem.

S6. Reproduction sequence and interpretation

Install the Python dependencies recorded in requirements_revision.txt. Obtain the original released inputs and download the raw files to revision_sources using the exact URLs in source_manifest.json. Preserve the raw snapshots and compare their checksums before recomputation. Run population aggregation, GHS comparison, topology audit, exact p-median audit, candidate-weight audit, road-accessibility processing, joint coverage, and the boundary check in the order specified in README_REVISION.md. The road graph has millions of vertices; sufficient memory and disk space are required. Download failure or different producer content must be reported, not silently replaced with simulated data.

The generated source counts and optimization outputs were checked in the present environment. A second independent clean-machine reproduction and matched empirical mobility validation have not been completed. The package supplies the materials for that reproduction rather than certifying an unperformed independent run.

S7 Extension executed on 7 October 2026

repeated_joint_stress_metadata.json

{
 "total_runs": 108,
 "five_stochastic_realizations": [
  20261007,
  20261008,
  20261009,
  20261010,
  20261011
 ],

```
  "baseline_seed": 0,
  "roads": "main classes",
  "grid_km": 5,
  "connector_km": 5,
  "threshold_km": [
    50,
    100,
    150
  ],
  "removal": "independent 1% feature deletion in each realization",
  "coordinates": "250 m fixed displacement with independently drawn azimuth; seed+100000",
  "limits": [
    "five realizations do not estimate tails",
    "uncalibrated stress magnitudes",
    "range/std are scenario descriptors, not statistical confidence",
    "joint deletion+coordinate perturbation cannot separately attribute their effects",
    "no empirical validation"
  ]
}
```


CANDIDATE_GRID_ENVIRONMENT_METADATA.JSON


```
{
  "boundary": "geoBoundaries common ten-province mask",
  "grid_origin": "EPSG6933 global zero; centres=(index+0.5)*step",
  "grid_km": [
    10,
    25
  ],
  "protected_database": "UNEP-WCMC and IUCN WDPCA October 2026 protected-area view, site_type PA",
  "polygon_records_cod": 59,
  "point_records_cod": 1,
  "included_status": [
    "Designated",
    "Inscribed",
    "Established"
  ],
  "proposed_records": "not included",
  "masked_modes": [
    "candidate centre within polygon",
    "full square cell intersects polygon or reported-area point buffer"
  ],
  "forest_modes": [
    "strict exclude WorldCover10",
    "allow WorldCover10 solely as upper-bound sensitivity; no inference of forest degradation"
  ],
  "worldcover": "ESA WorldCover2021v200 native point samples",
  "weights": "assumed six-variable reference vs uniform; no expert elicitation",
  "unknown_landcover": 0,
  "errors": [],
  "limits": [
    "no slope/parcel audit for new grid",
    "no forest integrity class",
    "tree-allowed candidates are not recommendations",
    "point buffers approximate reported area; not legal boundary",
    "population potential is a kernel, not observed flows",
    "no tenure or capacity validation",
    "protected polygon coordinates and attributes excluded from public package"
  ]
}
```

S8 Secondary observed traffic values

JICA Table 2.3.18, printed p. 2-94, reports observed screen-line counts of 311,333 PCU/day for SL A and 335,747 for SL B. Its calibrated model estimates are 287,878 and 316,870. Calculated relative differences are −7.533734% and −5.622388%. These are JICA model comparisons only. They are not predictions or validation errors of our hub-coverage model. The main report and appendices were inspected; raw GPS trajectories and a compatible regional OD target were not obtained.

S9 Protected-area inputs and redistribution

UNEP-WCMC and IUCN (2026), Protected Planet: The World Database on Protected and Conserved Areas (WDPCA), October 2026, Cambridge, UK: UNEP-WCMC and IUCN, https://www.protectedplanet.net/. The public protected-area view is filtered by iso3 containing COD, outSR=4326. It returns 59 polygon records and one point record without transfer-limit truncation. Designated, Inscribed and Established records enter the mask. Polygon union handles overlapping designations. Full-cell and centre tests are distinguished; reported-area point buffers are hypothetical circular screening footprints. Raw geometries and source attributes must not be included in the distributed revision archive. Obtain data from the producer under the current terms. Access date: 7 October 2026.

SHA-256 source polygon response: d7a8b8ef6b45e01117ab4693a3b5dccdf00f0982f1076ef3c815ef70dfa7a1ac. Point response: 3435b3c82dd5f341fa1be38a9044bc20ec8e5e6a221f010725409795b3142f3e. File hashes preserve the response snapshot; they do not imply redistribution permission.

Additional variance and settlement analyses

An exact functional ANOVA decomposes coverage variance on the complete 96-configuration factorial table separately for p = 5, 10, and 15. The six categorical factors are population product, demand-grid resolution, road-class set, connector distance, deletion level, and coordinate-displacement level. Each enumerated level is assigned equal independent probability for this calculation only. For each factor subset, its variance component equals the variance of its conditional cell means minus the components of all proper nonempty subsets. Dividing by total variance gives a first-order share for each singleton; summing all components containing that factor gives its total-effect share. All 63 nonempty components are included and their sum is checked against total variance. This is finite-design attribution under a declared scenario measure, not a calibrated error distribution or a continuous-parameter Monte Carlo Sobol estimate [27], [28].

Morphological settlement footprints are extracted from the native 2025 GHS-SMOD R2023A V2.0 one-kilometre Mollweide raster [14]–[18]. Published codes 30 (urban centre), 23 (dense urban cluster), and 22 (semi-dense urban cluster) are labelled separately using four-connected cells. The complete COD administrative union is applied by cell-centre inclusion before labelling; study-region selection follows labelling, avoiding artificial splits at provincial boundaries. Producer classification and smoothing are retained. Native, aligned GHS-POP cells are summed over each component. These derived classified-cell polygons are not official UCDB identifiers, functional urban areas, or administrative city populations. Hub membership requires its point to lie in a component; unmatched points are retained. Rural demand is not discarded.

Verification: each factorial table contains exactly 96 unique configurations, with every factor combination represented. Nonnegative ANOVA components sum to the observed variance within floating-point tolerance for all three hub counts. Inputs remain the already-computed 288 exact covering runs; no new covering optimizations are claimed. Population and settlement source archives are downloaded and checked against the retained SHA-256 manifest. Native population and SMOD window transforms, dimensions, and coordinate systems are verified before summation.

Reproduction files: analyze_variance.py; delineate_ghsl.py; download.py; variance_factor_indices.csv; variance_components_p5.csv, p10.csv, p15.csv; ghsl_cluster_inventory.csv; existing_hubs_cluster_membership.csv; ghsl_morphological_clusters.gpkg; and metadata JSON files. The GeoPackage is directly usable in QGIS. Derived component identifiers depend on the stated country mask and labelling order; they are local identifiers, not GHSL producer IDs.

Outstanding evidence: no compatible regional flow, journey, access-use, or field observation has been matched to the covering result. No expert has been recruited or interviewed. The consultation guide is prospective. Secondary traffic counts and modelled population products must not be described as validation of regional population coverage.